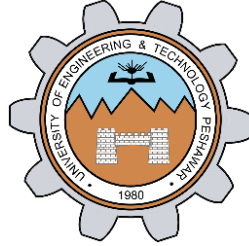


Introduction to Cisco Packet Tracer

LAB # 07



Spring 2024

CSE-303L Data Communication & Networks Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Yasir Saleem Afridi

Date:

3rd June 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

CSE 303L: Data Communication and Computer Networks

Credit Hours: 1

Demonstration of Concepts	Poor (Does not meet expectation (1)) The student failed to demonstrate a clear understanding of the assignment concepts	Fair (Meet Expectation (2-3)) The student demonstrated a clear understanding of some of the assignment concepts	Good (Exceeds Expectation (4-5)) The student demonstrated a clear understanding of the assignment concepts	Score 30%
Accuracy	The student mis-configured enough network settings that the lab computer couldn't function properly on the network	The student configured enough network settings that the lab computer partially functioned on the network	The student configured the network settings that the lab computer fully functioned on the network	30%
Following Directions	The student clearly failed to follow the verbal and written instructions to successfully complete the lab	The student failed to follow the some of the verbal and written instructions to successfully complete all requirements of the lab	The student followed the verbal and written instructions to successfully complete requirements of the lab	20%
Time Utilization	The student failed to complete even part of the lab in the allotted amount of time	The student failed to complete the entire lab in the allotted amount of time	The student completed the lab in its entirety in the al	20%

Objectives of the Lab:

This lab aims to introduce Cisco Packet Tracer. Some specific topics covered in this lab are

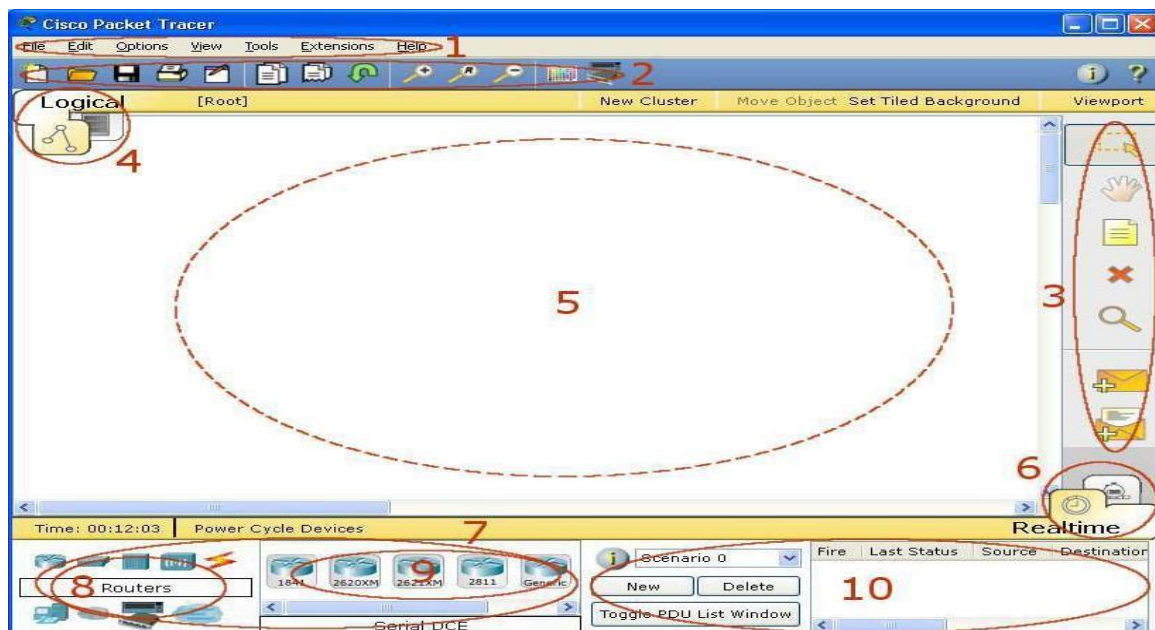
- Downloading Cisco Packet Tracer from NetAcad
 - Installing Packet Tracer • Cisco Packet Tracer Overview
 - Creating Devices
 - Adding Modules
 - Making Connections
 - Creating Networks
-

Introduction

Cisco Packet Tracer is an innovative network simulation and visualization tool. This free software helps you to practice your network configuration and troubleshooting skills via your desktop computer or an Android or iOS based mobile device. Packet Tracer is available for both the Linux and Windows desktop environments.

With Packet Tracer you can choose to build a network from scratch, use a pre-built sample network, or complete classroom lab assignments. Packet Tracer allows you to easily explore how data traverses your network. Packet Tracer provides an easy way to design and build networks of varying sizes without expensive lab equipment.

1. Downloading
2. Installing Cisco Packet Tracer
3. Cisco Packet Tracer overview



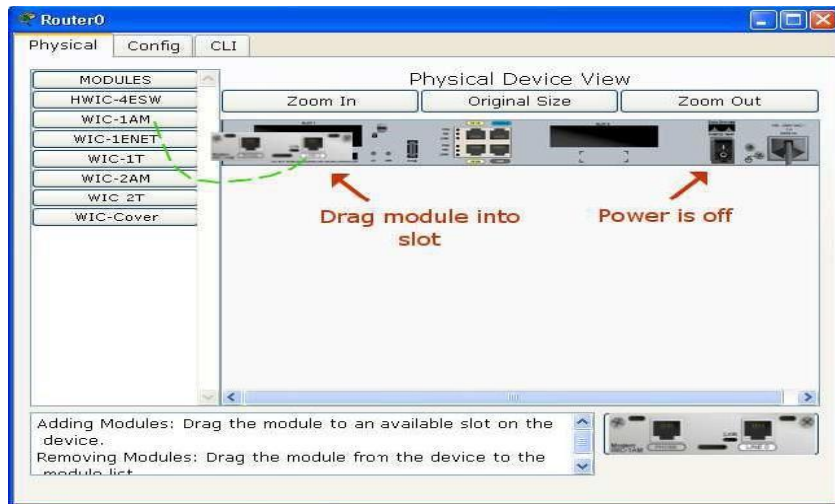
1	Menu Bar	This bar provides the File, Edit, Options, View, Tools, Extensions, and Help menus. You will find basic commands such as Open, Save, Print, and Preferences in these menus. You will also be able to access the Activity Wizard from the Extensions menu.
2	Main Tool Bar	This bar provides shortcut icons to the File and Edit menu commands. This bar also provides buttons for Zoom, the drawing Palette, and the Device Template Manager. On the right, you will also find the Network Information button, which you can use to enter a description for the current network (or any text you wish to include).
3	Common Tools Bar	This bar provides access to these commonly used workspace tools: Select, Move Layout, Place Note, Delete, Inspect, Add Simple PDU, and Add Complex PDU. See "Workspace Basics" for more information.
4	Logical/Physical Workspace and Navigation Bar	You can toggle between the Physical Workspace and the Logical Workspace with the tabs on this bar. In Logical Workspace, this bar also allows you to navigate through levels of a cluster, create a new New Cluster, Move Object, Set Tiled Background, and Viewport. In Physical Workspace, this bar allows you to navigate through physical locations, create a New City, create a New Building, create a New Closet, Move Object, apply Grid to the background, Set Background, and go to the Working Closet.
5	Workspace	This area is where you will create your network, watch simulations, and view many kinds of information and statistics.
6	Realtime/Simulation Bar	You can toggle between Realtime Mode and Simulation Mode with the tabs on this bar. This bar also provides buttons to Power Cycle Devices as well as the Play Control buttons and the Event List toggle button in Simulation Mode. Also, it contains a clock that displays the relative Time in Realtime Mode and Simulation Mode.
7	Network Component Box	This box is where you choose devices and connections to put into the workspace. It contains the Device-Type Selection Box and the Device-Specific Selection Box.
8	Device-Type Selection Box	This box contains the type of devices and connections available in Packet Tracer 5.1. The Device-Specific Selection

		Box will change depending on which type of device you choose.
9	Device-Specific Selection Box	This box is where you choose specifically which devices you want to put in your network and which connections to make.
10	User Created Packet Window*	This window manages the packets you put in the network during simulation scenarios. See the "Simulation Mode" section for more details.

Creating Devices

- Choose a device type from the **Device-Type Selection** box
- Click on the desired device model from the **Device-Specific Selection** box
- Click on a location in the workspace to put your device in that location
- If you want to cancel your selection, press the **Cancel** icon for that device
- Alternatively, you can click and drag a device from the **Device-Specific Selection** box onto the workspace
- You can also click and drag a device directly from the **Device-Type Selection** box and a default device model will be chosen for you

Adding Modules



- Click on a device to bring up its configuration window.
- By default, you will be in the **Physical Device View** subpanel of the device.
- You can browse (by clicking) through the list of modules and read their description in the information box at the bottom.

- d. When you have found the module you want to add, simply drag it from the list into a compatible bay on the device picture.
- e. You can remove a module by dragging it from the device back into the list.

Making Connections

- a. To make a connection between two devices, first click the **Connections** icon from the **Device-Type Selection** box to bring up the list of available connections.
- b. Then click the appropriate cable type.
- c. The mouse pointer will change into a "connection" cursor.
- d. Click on the first device and choose an appropriate interface to which to connect.
- e. Then click on the second device and do the same.
- f. A connection cable will appear between the two devices, along with link lights showing the link status on each end (for interfaces that have link lights).

Creating Networks

Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command line interface.

Lab Report Tasks

Task 1: Installation and Overview

- **Download and Install Cisco Packet Tracer:** Document the steps taken to download and install Packet Tracer from NetAcad.
- **Overview of Cisco Packet Tracer Interface:** Provide a brief description of the following components in Packet Tracer:
 - 1. Menu Bar
 - 2. Main Tool Bar
 - 3. Common Tools Bar
 - 4. Logical/Physical Workspace and Navigation Bar
 - 5. Workspace

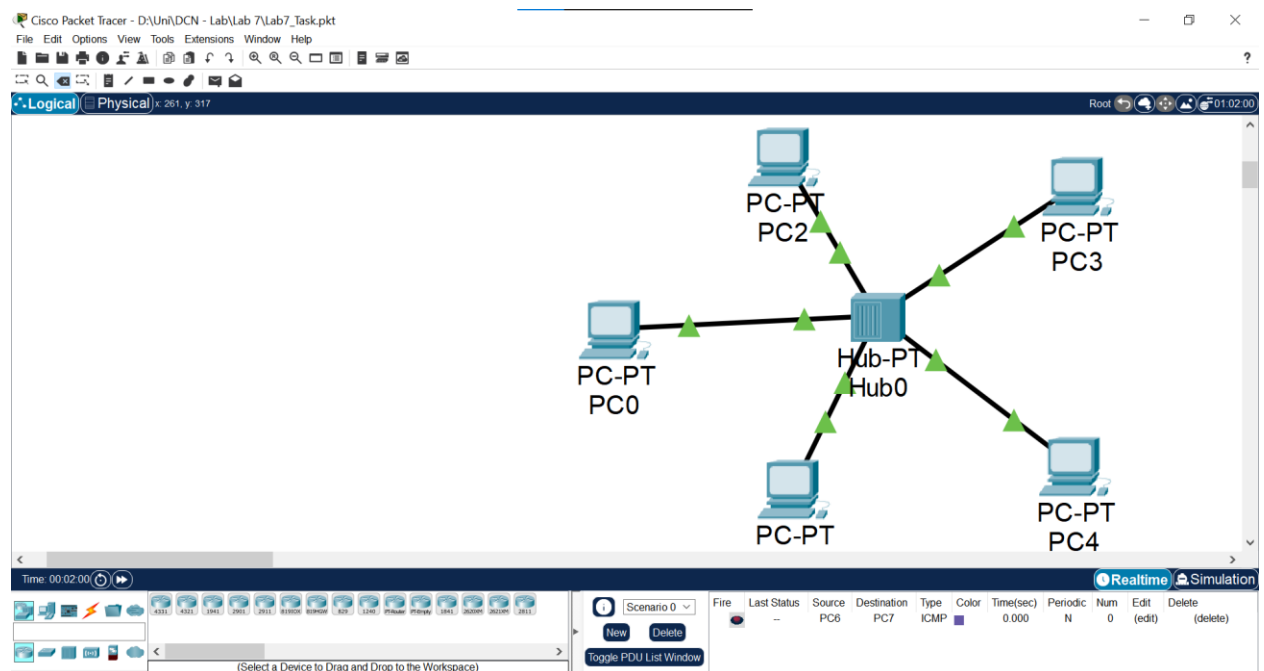
6. Realtime/Simulation Bar
7. Network Component Box
8. Device-Type Selection Box
9. Device-Specific Selection Box
10. User Created Packet Window

Task 2: Creating Devices

Device Creation:

We can add devices to our workspace from the bottom tab. Here we have a lot of choices and types of devices. Some of the major types are:

Network Devices, End Devices, Components



Task 3: Making Connections

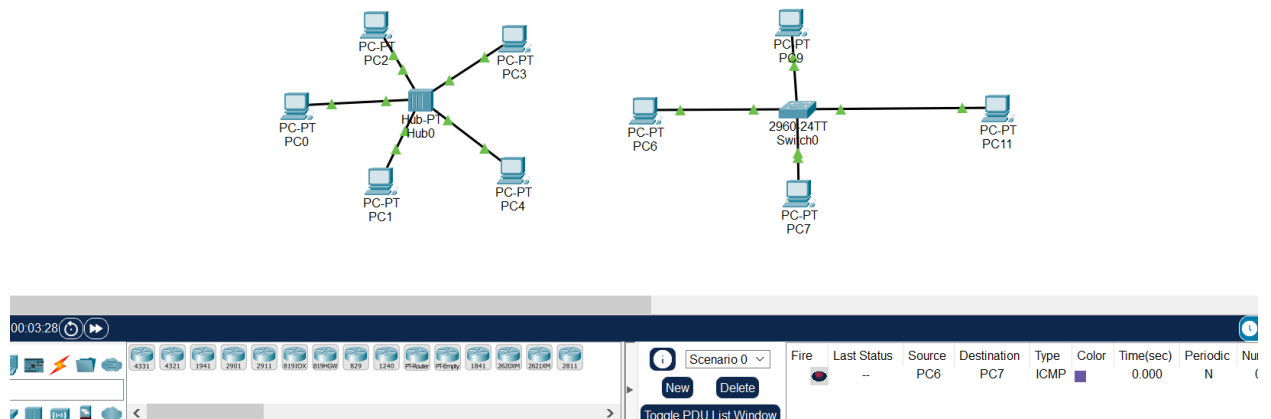
Connection Setup:

To make connections between devices using different cable types in Cisco Packet Tracer, first, click the "Connections" icon (lightning bolt) to open the cable options. Select the appropriate cable type based on the devices you are connecting:

1. Copper Straight-Through Cable: Connects different device types (e.g., PC to switch).
2. Copper Crossover Cable: Connects similar device types (e.g., switch to switch).
3. Console Cable: Connects a PC to the console port of a router or switch for configuration.

Click on the first device, select the appropriate port, and then click on the second device to complete the connection. Verify connectivity by observing the link status lights.

- **Screenshot:**

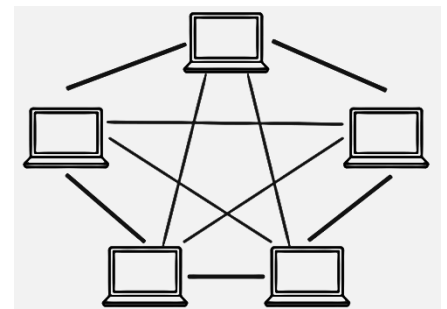


Task 4: Creating Networks

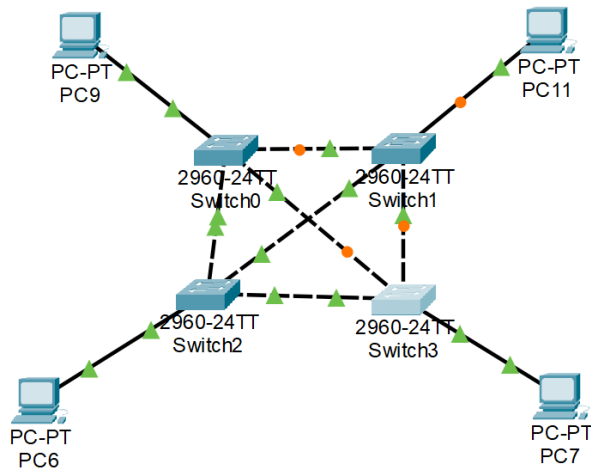
- **Network Creation:** Demonstrate creating a simple network topology with at least three different devices.
- **Network Description:**
I have created a network based on mesh topology.

Mesh Topology:

Mesh topology is a network configuration where devices are interconnected in a decentralized manner. Instead of relying on a central hub or switch, each device connects directly to multiple other devices, forming a mesh-like structure. It allows for multiple paths between devices, enhancing redundancy and fault tolerance.



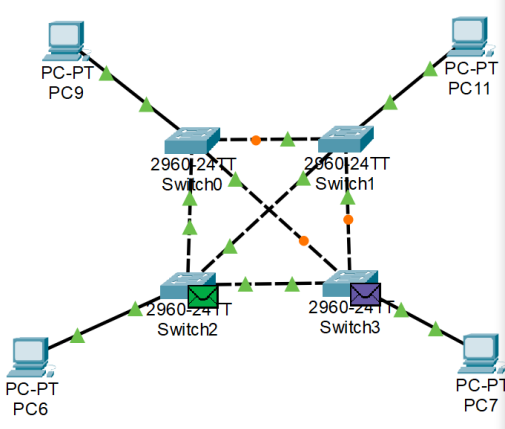
- **Screenshot:**



Task 5: Connectivity Testing

- **Ping Command:** Explain the steps to use the ping command to test connectivity between devices in Packet Tracer.
- **Testing:**
 - 1) First of all, I assigned IP addresses to all four end devices using Static IP method in IP Configuration.
 - 2) I followed this IP addressing scheme in this case only.
PC6(192.168.1.10), PC9(192.168.1.11), PC11(192.168.1.12) and PC7(192.168.1.13).
 - 3) Then I opened Command Prompt in the Desktop and entered ping 192.168.1.10 and 3 packets were successfully sent to PC6.

- **Screenshot:**



Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=35ms TTL=128
Reply from 192.168.1.10: bytes=32 time=2ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 35ms, Average = 9ms

C:\>
```